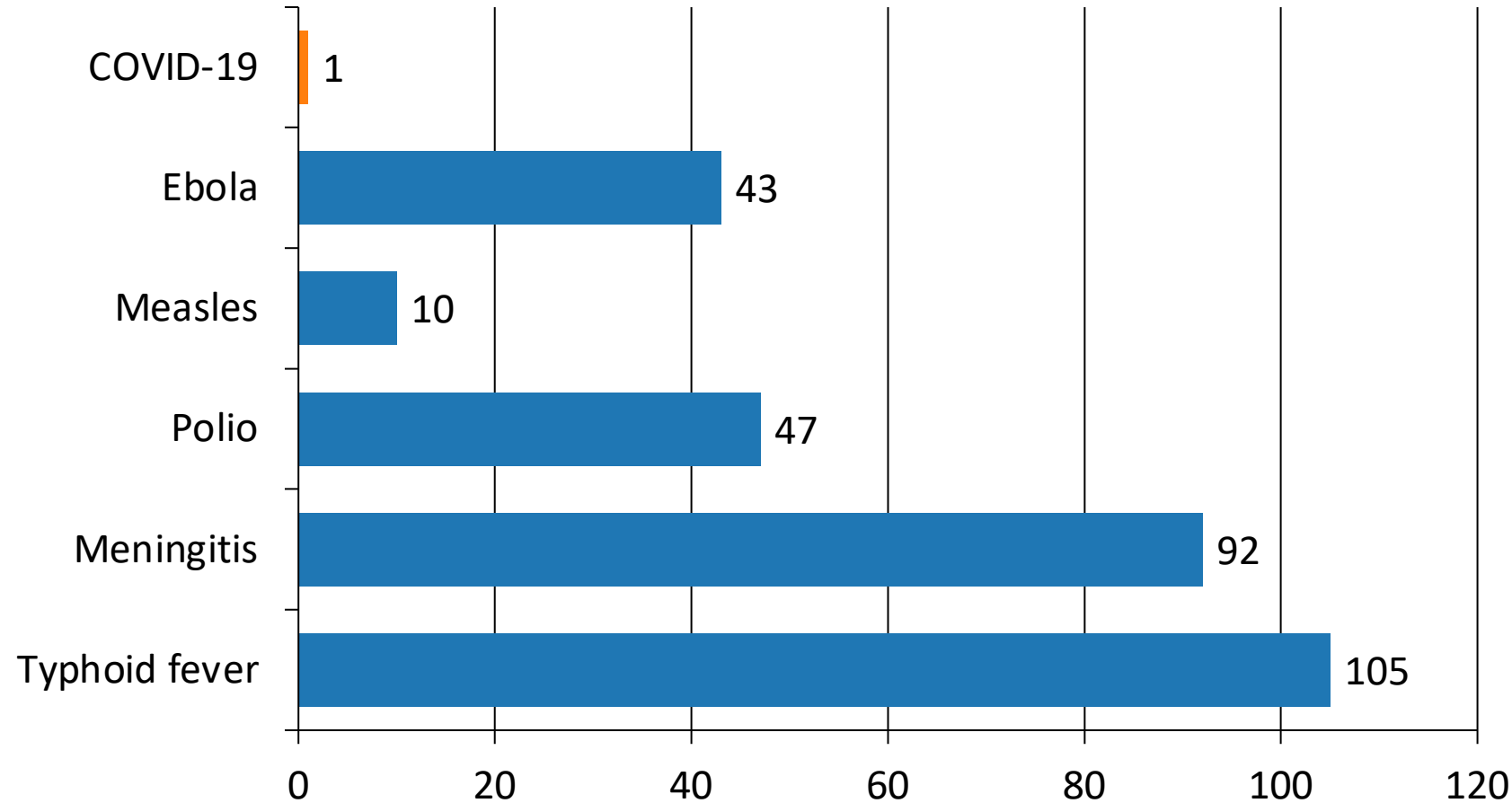


The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview (1796–2020) for public health
professionals and policy analysts

Source: Our World in Data — Vaccination

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



COVID-19: <1 year (2020 → 2020)

Pathogen identification → vaccine licensure gaps have shortened dramatically. Example: typhoid fever took 105 years (1884 → 1989). Modern tools (genome sequencing, mRNA technology) enable faster design.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-vaccine cases (per million/yr)	Post-vaccine cases (per million/yr)	Case reduction	Death reduction
Diphtheria	158	0	100%	100%
Smallpox	250	0	100%	100%
Acute polio	141	0	100%	100%
Measles	3,044	0.2	99.99%	100%
Pneumococcal	233	139	40.5%	31.3%

Several US vaccine-preventable diseases achieved 100% elimination of both cases and deaths.

Measles fell to 0.2 cases per million/year (99.99% reduction).

Remaining burden persists for pneumococcal disease (40.5% case reduction; 31.3% death reduction).

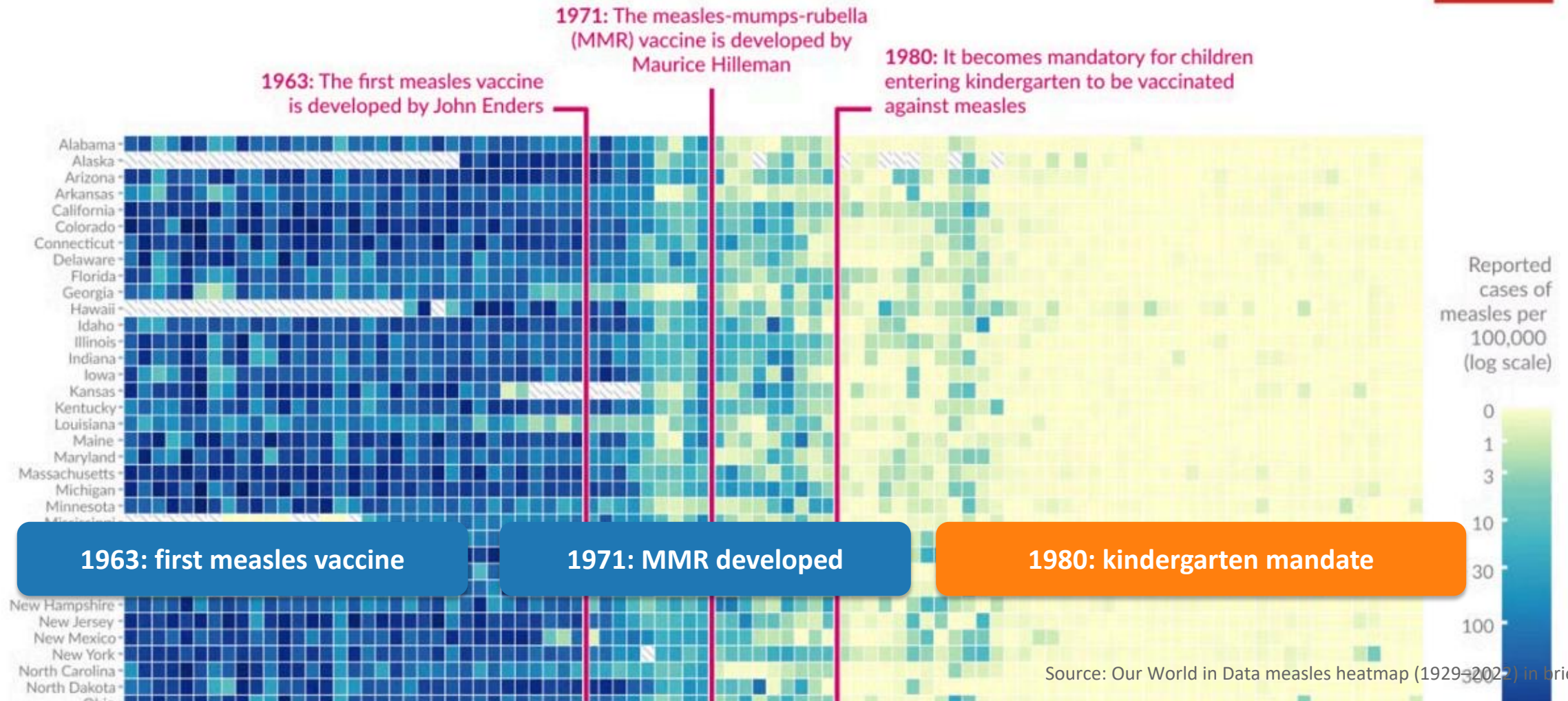
Measles Cases Collapsed Across All 50

US States After Vaccine Mandates

Before 1963, nearly every state experienced 100–1,000+ cases per 100,000; after 1980, near-zero with sporadic outbreaks.

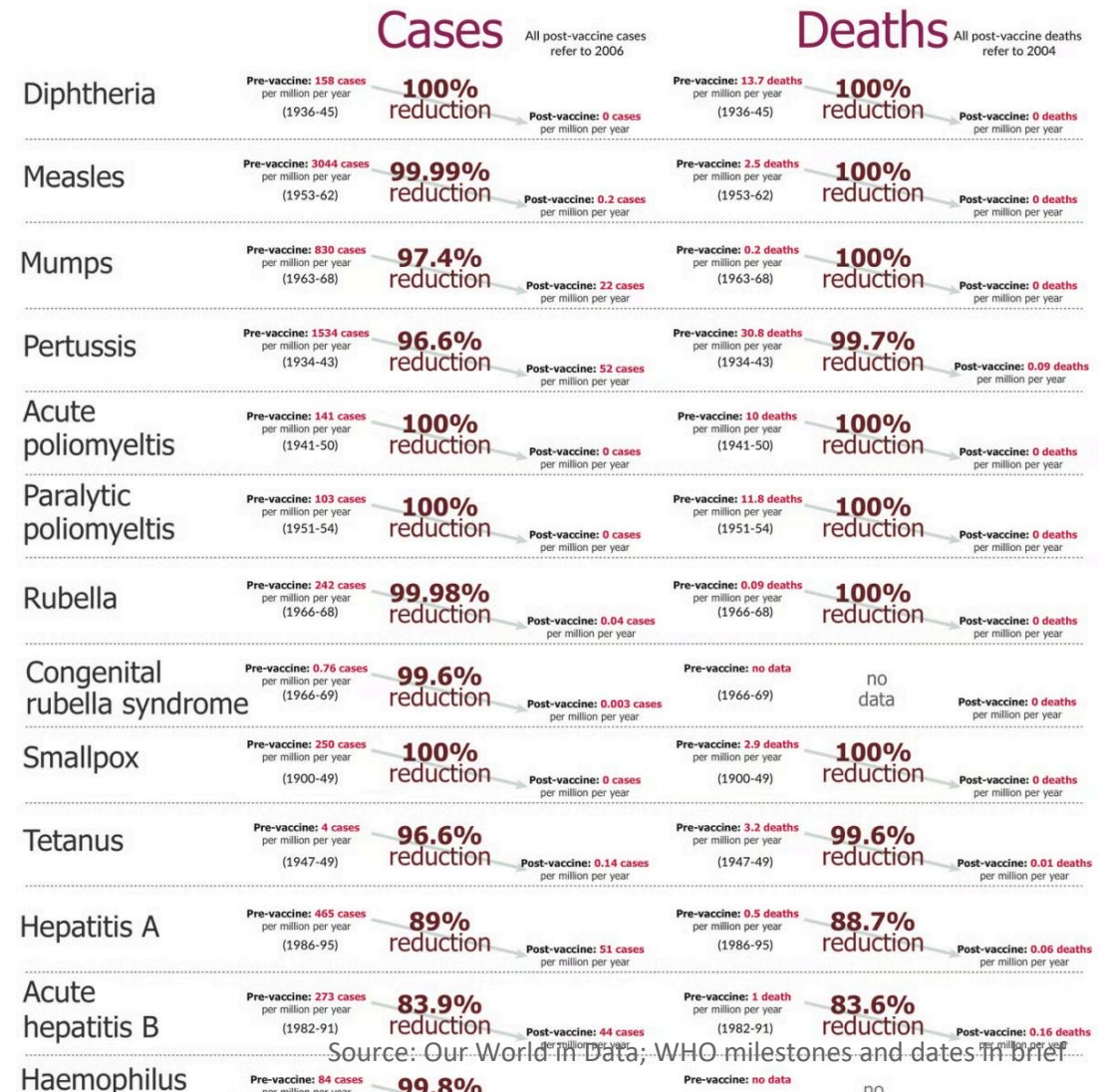
Vaccines reduced measles cases across US states

Our World
in Data



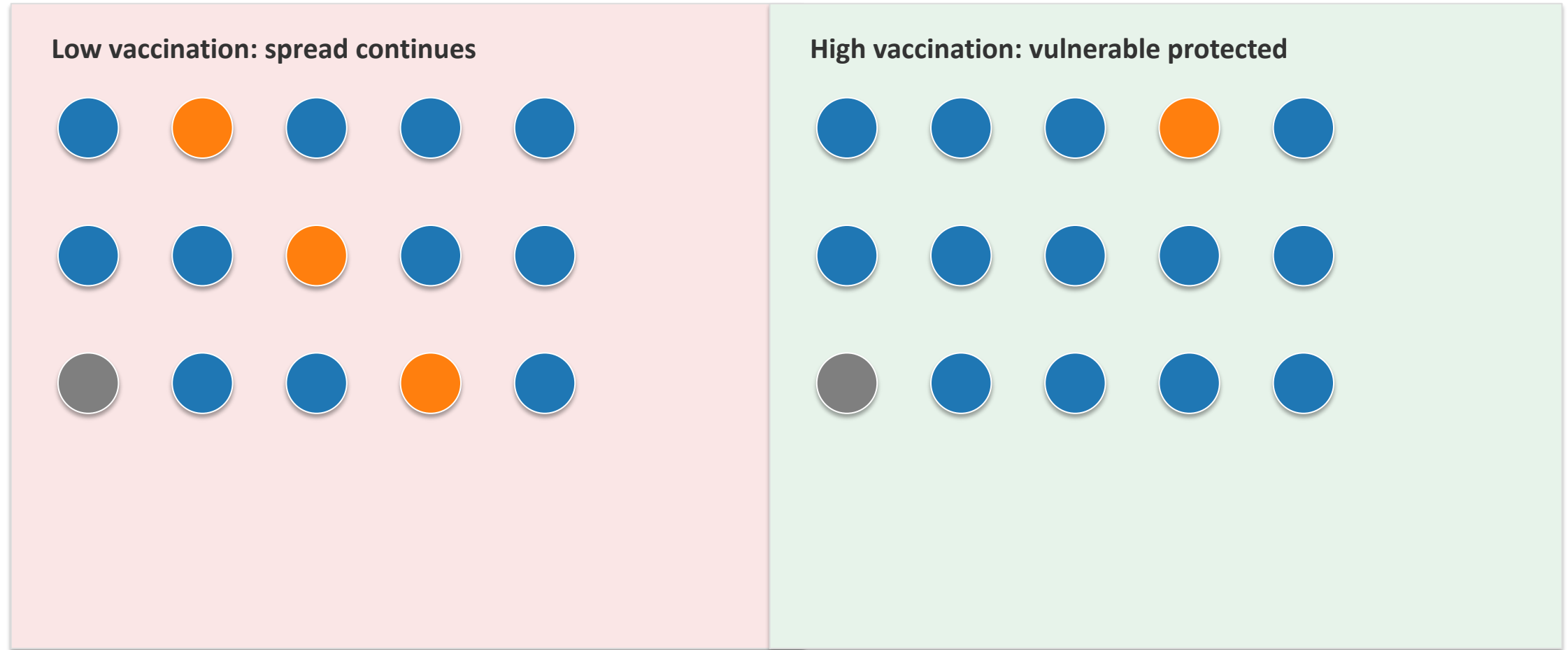
Smallpox: The Only Human Disease Eradicated Through Vaccination

- **Key timeline**
- 1959: WHO passed a resolution to eradicate smallpox
- 1967: intensified global campaign launched
- 1977: last natural case occurred in Somalia
- **1980: smallpox declared eradicated worldwide**
- **Why eradication was feasible**
- Infected only humans
- Symptoms were distinguishable
- Vaccine was cheap and effective



Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Vaccines protect communities: when enough people are vaccinated, pathogens can't spread effectively — shielding those who cannot be

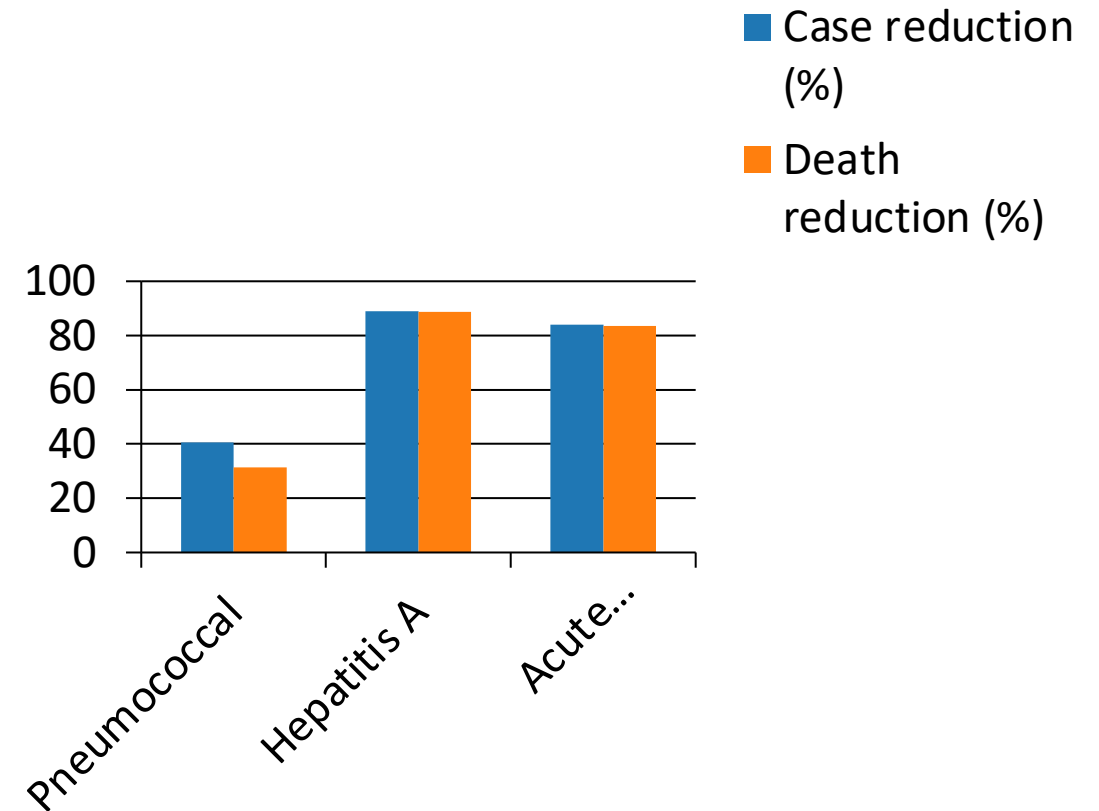


Blue = vaccinated Orange = infected Gray = cannot be vaccinated (e.g., newborns, immunocompromised, chemotherapy patients)

Source: herd immunity description in brief (Our World in Data)

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

- Incomplete disease control (US reductions, per million/year)
- **Pneumococcal: 40.5% case reduction; 31.3% death reduction**
- Hepatitis A: 89% case reduction; 88.7% death reduction
- Acute hepatitis B: 83.9% case reduction; 83.6% death reduction
- **System challenges (global)**
- Not everyone has access to vaccines
- Supply shortages put children at risk
- Poor scientific understanding can reduce trust and uptake



Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- Vaccines achieved 100% elimination of cases and deaths in the US for diphtheria, smallpox, and acute polio.
- Development timelines collapsed from 105 years (typhoid) to <1 year (COVID-19).
- Measles fell to near zero across all 50 states after the 1980 kindergarten mandate, with sporadic outbreaks.
- Herd immunity protects newborns, immunocompromised individuals, and chemotherapy patients who cannot be vaccinated.
- Coverage and access gaps persist (e.g., pneumococcal 40.5% case reduction).

Call to action: strengthen access, coverage, and trust — and sustain systems that deliver vaccines reliably.